Final Project Demonstration Expectations

The final project demonstration is during your regular lab section on either 4/20 Monday or 4/21 Tuesday.

You are expected to demonstrate your final solution to the problem of tracking the number of people using the demo door in the lab (for the people counting system) or different scenarios defined by each team for the team-defined project.

For the people counting project, your system will be evaluated with the following baseline and edge case scenarios. For the team-defined project, each team is responsible for defining baseline and edge case scenarios (get approval from the instructor before your final demonstration). **Your team will be assigned to one of two test doors in the lab. While one team is demonstrating its system, the next team must complete installing the system on the other door so that the demo can start immediately after the demo of the previous team.** We will try to assign each team to their preferred door, but it will not be guaranteed (i.e., will be randomly assigned) if the door preference is not balanced.

Baseline scenario for people counting systems

- Single person slow walk in / out, long (>= 3 sec) time gap between walk in / out
- Single person fast walk in / out, long (>= 3 sec) time gap between walk in / out
- Multiple people walk in or out all in the same direction with ~1.5m spacing between people
- Multiple people walk in or out in random directions without >=2 people passing the door at the same time

Edge cases for people counting systems

- Two people walk in or out in the same direction with no or small time gap between them when they pass the door (i.e., brushing shoulders)
- Two people walk in and out in the opposite direction with no or small time gap between them when they pass the door (i.e., brushing shoulders)
- One person walks part way in (or out) then walk out (or in)
- One person in a wheelchair enters the room
- One person in a wheelchair is being pushed by another person into the room
- People standing next to the doorway while another person walks in or out

- Other ad hoc situations depending on the team's specific design (i.e., scenario defined by the team to show the unique feature of the design)

Start your demo with a short (<1 min) system introduction presentation. During the presentation, explain how the system works and the power consumption of your system with an analysis of the expected lifetime (with and without power-saving features if you implemented any).

Rubric (total 20 pts)

System introduction presentation including power analysis (3 pts)

Completeness and readiness of the final system for the baseline scenario demo (9 pts)

Reliability and performance of the final system for the baseline and edge cases (8 pts)